UNIT-5

FOURIER TRANSFORMS

FORMULAE

1. *Fourier Transform of f(x) is* $F[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$

2. *The inversion formula* $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{-isx} ds$

3. *Fourier cosine Transform* $F_c [f(x)] = F_c(s) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos sx dx$

4. *Inversion formula* $f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_c(s) \cos sx ds$

5. *Fourier sine Transform (FST)* $F_s [f(x)] = F_s(s) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin sx dx$

6. *Inversion formula* $f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_s(s) \sin sx ds$

7. Parseval's Identity $\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |F(s)|^2 ds$

8. *Gamma function* $\Gamma n = \int_0^{\infty} x^{n-1} e^{-x} dx$, $\Gamma n+1 = n\Gamma n$ & $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

9. $\int_0^{\infty} e^{-ax} \cos bx dx = \frac{a}{a^2+b^2}$

10 $\int_0^{\infty} e^{-ax} \sin bx dx = \frac{b}{a^2+b^2}$

11. $\int_0^{\infty} \frac{\sin ax}{x} dx = \frac{\pi}{2}$

12. $\int_0^\infty e^{-x^2}dx = \frac{\sqrt{\pi}}{2}$ & $\int_{-\infty}^\infty e^{-x^2}dx = \sqrt{\pi}$

13. $\cos ax = \frac{e^{iax}+e^{-iax}}{2}$ & $\sin ax = \frac{e^{iax}-e^{-iax}}{2}$

ORKING RULE TO FIND THE FOURIER TRANSFORM

Step1: Write the FT formula.

Step2: Substitute given f(x) with their limits.

Step3: Expand e^{isx} *as cos sx + isin sx and use Even & odd property*

*Step4: Integra*te by using Bernoulli's formula then we get F(s)

WORKING RULE TO FIND THE INVERSE FOURIER TRANSFORM

Step1: Write the Inverse FT formula

Step2: Sub f(x) & F(s) with limit $-\infty,\infty$ *in the formula*

Step3: Expand e^{-isx} *as cos sx -isin sx and equate real part*

Step4: Simplify we get result

WORKING RULE FOR PARSEVAL'S IDENTITY

If F(s) is the Fourier transform of f(x) then

$\int_{-\infty}^{\infty}\left|f(x)\right|^2 dx = \int_{-\infty}^{\infty}\left|F(s)\right|^2 ds$ is known as Parseval's identity.

Step1: Sub f(x) & F(s) With their limits in the above formula

Step2: Simplify we get result

WORKING RULE TO FIND FCT

Step1: Write the FCT formula & Sub f(x) with its limit in the formula

Step2: Simplify, we get $F_C(S)$

WORKING RULE TO FIND INVERSE FCT

Step1: Write the inverse FCT formula & Sub $F_C(S)$ *with its limit in the formula*

Step2: Simplify, we get f(x)

WORKING RULE TO FIND FST

Step1: Write the FST formula & Sub f(x) with its limit in the formula

Step2: Simplify, we get $F_S(S)$

WORKING RULE TO FIND INVERSE FCT

Step1: Write the inverse FST formula & Sub $F_s(S)$ *with limit in the formula*

Step2: Simplify, we get f(x)

WORKING RULE FOR f(x) = e^{-ax}

Step:1 First we follow the above FCT & FST working rule and then we get this result

$F_c(e^{-ax}) = \sqrt{\frac{2}{\pi}}\frac{a}{a^2+s^2}$

By Inversion formula,

$$\int_0^\infty \frac{\cos sx}{a^2+s^2}\,ds = \frac{\pi}{2a}e^{-ax}$$

$F_s(e^{-ax}) = \sqrt{\frac{2}{\pi}}\frac{s}{a^2+s^2}$

By Inversion formula,

$$\int_0^\infty \frac{s}{a^2+s^2}\sin sx\,ds = \frac{\pi}{2}e^{-ax}$$

TYPE-I : *If problems of the form* *i)* $\frac{x}{x^2+a^2}$ *ii)* $\frac{1}{x^2+a^2}$ **,** *then use Inversion formula*

TYPE-II: *If problems of the form i)* $\int_0^\infty \frac{x^2}{(x^2+a^2)^2}dx$ *ii)* $\int_0^\infty \frac{dx}{(x^2+a^2)^2}$**,** then use Parseval's Identity

TYPE-III

$\int_0^\infty \frac{dx}{(x^2+a^2)(x^2+b^2)}$ *, then use* $\int_0^\infty f(x)g(x)dx = \int_0^\infty F_C[f(x)]\,F_C[g(x)]\,dx$

FOURIER TRANSFORMS

1. State Fourier Integral Theorem.

<u>Answer:</u>

If $f(x)$ *is piece wise continuously differentiable and absolutely on* $-\infty,\infty$ *then,*

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f\,t\,e^{\,i\,(x-t\,s)}\,dt\,ds.$$

2. **State and prove Modulation theorem.** $F\left[f(x)\cos ax\right]=\frac{1}{2}\left[F(s+a)+F(s-a)\right]$ **Proof:**

$$F\left[f(x)\cos ax\right]=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x)\cos ax\, e^{isx}dx$$

$$=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x)\left(\frac{e^{iax}+e^{-iax}}{2}\right)e^{isx}dx$$

$$=\frac{1}{2}\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x)\,e^{i(s+a)x}dx+\frac{1}{2}\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x)\,e^{i(s-a)x}dx$$

$$=\frac{1}{2}F(s+a)+\frac{1}{2}F(s-a)$$

$$F\left[f(x)\cos ax\right]=\frac{1}{2}\left[F(s+a)+F(s-a)\right]$$

3. **State Parseval's Identity.**

Answer:

If $F(s)$ is a Fourier transform of $f(x)$, then

$$\int_{-\infty}^{\infty}\left|F(s)\right|^2 ds=\int_{-\infty}^{\infty}\left|f(x)\right|^2 dx$$

4. **State Convolution theorem.**

Answer:

The Fourier transform of Convolution of $f(x)$ and $g(x)$ is the product of their Fourier transforms.

$$F(f*g)=F(s)G(s)$$

5. **State and prove Change of scale of property.**

Answer:

If $F(s) = F[f(x)]$, then $F[f(ax)] = \frac{1}{a} F\left(s/a\right)$

$$F[f(ax)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(ax) e^{isx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{i\left(s/a\right)t} \frac{dt}{a}; \qquad \text{where } t = ax$$

$$F[f(ax)] = \frac{1}{a} F\left(s/a\right)$$

6. **Prove that if F[f(x)] = F(s) then** $F[x^n f(x)] = (-i)^n \frac{d^n}{ds^n} F(s)$

Answer:

$$F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$$

Diff w.r.t *s* '*n*' times

$$\frac{d^n}{ds^n} F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) (ix)^n e^{isx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) (i)^n (x)^n e^{isx} dx$$

$$\frac{1}{(i)^n} \frac{d^n}{ds^n} F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (x)^n f(x) e^{isx} dx$$

$$(-i)^n \frac{d^n}{ds^n} F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (x)^n f(x) e^{isx} dx$$

$$F\left[x^n f(x)\right] = (-i)^n \frac{d^n}{ds^n} F(s)$$

7. **Solve for f(x) from the integral equation** $\int_0^{\infty} f(x) \cos sx\, dx = e^{-s}$

Answer:

$$\int_0^\infty f(x)\cos sx\,dx = e^{-s}$$

$$F_c\left[f\ x\right] = \sqrt{\frac{2}{\pi}}\int_0^\infty f\ x\cos sx\,dx$$

$$F_c\left[f\ x\right] = \sqrt{\frac{2}{\pi}}\, e^{\,s-}$$

$$f(x) = \sqrt{\frac{2}{\pi}}\int_0^\infty F_c\left[f\ x\right]\cos sx\,ds$$

$$= \sqrt{\frac{2}{\pi}}\int_0^\infty \sqrt{\frac{2}{\pi}}\, e^{\,s-}\cos sx\,ds$$

$$= \frac{2}{\pi}\int_0^\infty e^{-s}\cos sx\,ds \qquad = \frac{2}{\pi}\left[\frac{1}{x^2+1}\right]$$

$$\int_0^\infty e^{-ax}\cos bx\,dx = \frac{a}{a^2+b^2}$$

$$\therefore\ a = 1,\ b = x$$

8. Find the complex Fourier Transform of $f(x) = \begin{cases} 1 & |x| < a \\ 0 & |x| > a > 0 \end{cases}$

Answer:

$$F\left[f\ x\right] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f\ x\, e^{isx}\,dx$$

$$|x| < a;\ -a < x < a$$

$$F\left[f\ \ x\right] = \frac{1}{\sqrt{2\pi}}\int_{-a}^{a} 1\, e^{isx} dx$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-a}^{a} (\cos sx + i\sin sx) dx$$

$$= \frac{2}{\sqrt{2\pi}}\int_{0}^{a} (\cos sx) dx \ = \frac{2}{\sqrt{2\pi}}\left(\frac{\sin sx}{s}\right)_0^a$$

$$= \sqrt{\frac{2}{\pi}}\left(\frac{\sin as}{s}\right)$$

[Use even and odd property second term become zero]

9. Find the complex Fourier Transform of $f(x) = \begin{cases} x & |x| < a \\ 0 & |x| > a > 0 \end{cases}$

Answer:

$$F\left[f\ \ x\right] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f\ \ x\, e^{isx} dx$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-a}^{a} x e^{isx} dx \qquad \because\ |x| < a;\ -a < x < a$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-a}^{a} x(\cos sx + i\sin sx) dx$$

$$= \frac{2}{\sqrt{2\pi}}\int_{0}^{a} (x(i\sin sx) dx \ = \frac{2i}{\sqrt{2\pi}}\left[x\left(\frac{-\cos sx}{s}\right) - (1)\left(\frac{-\sin sx}{s^2}\right)\right]_0^a$$

$$= i\sqrt{\frac{2}{\pi}}\left[\frac{-as\cos as + \sin as}{s^2}\right]$$

[Use even and odd property first term become zero]

10. Write Fourier Transform pair.

Answer:

If $f(x)$ *is defined in* $(-\infty,\infty)$ *, then its Fourier transform is defined as*

$$F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\, e^{isx} dx$$

If $F(s)$ *is an Fourier transform of* $f(x)$ *, then at every point of Continuity of* $f(x)$ *, we have* $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{-isx} ds$.

11. Find the Fourier cosine Transform of f(x) = e[-x]

Answer:

$$F_c\left[f(x)\right] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos sx dx$$

$$F_c\left[e^{-x}\right] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-x} \cos sx dx \qquad \int_0^{\infty} e^{-ax} \cos bx dx = \frac{a}{a^2+b^2}$$

$$F_c\left[e^{-x}\right] = \sqrt{\frac{2}{\pi}} \frac{1}{s^2+1}$$

12. Find the Fourier Transform of $f(x) = \begin{cases} e^{imx} & , a < x < b \\ 0, & otherwise \end{cases}$

Answer:

$$F\left[f(x)\right]=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x)\,e^{isx}dx$$

$$=\frac{1}{\sqrt{2\pi}}\int_a^b e^{imx}e^{isx}dx \quad =\frac{1}{\sqrt{2\pi}}\int_a^b e^{i(m+s)x}\,dx$$

$$=\frac{1}{\sqrt{2\pi}}\left[\frac{e^{i(m+s)x}}{i(m+s)}\right]_a^b \quad =\frac{1}{\sqrt{2\pi}}\frac{1}{i(m+s)}\left[e^{i(m+s)b}-e^{i(m+s)a}\right]$$

13. Find the Fourier sine Transform of $\frac{1}{x}$.

Answer:

$$F_s\left[f(x)\right]=\sqrt{\frac{2}{\pi}}\int_0^{\infty} f(x)\sin sx\,dx$$

$$=\sqrt{\frac{2}{\pi}}\int_0^{\infty}\frac{\sin sx}{x}\,dx \;=\sqrt{\frac{2}{\pi}}\;\frac{\pi}{2}$$

$$F_s\left[\frac{1}{x}\right]=\sqrt{\frac{\pi}{2}}$$

14. **Find the Fourier sine transform of** e^{x-}

Answer:

$$F_s\left[f(x)\right]=\sqrt{\frac{2}{\pi}}\int_0^{\infty} f(x)\sin sx\,dx$$

$$F_s\left[e^{-x}\right]=\sqrt{\frac{2}{\pi}}\int_0^{\infty} e^{-x}\sin sx\,dx$$

$$\because \int_0^{\infty} e^{-ax}\sin bx\,dx=\frac{b}{a^2+b^2}$$

$$F_s\left[e^{-x}\right]=\sqrt{\frac{2}{\pi}}\frac{s}{s^2+1}$$

15. **Find the Fourier cosine transform of** $e^{-2x}+2e^{-x}$

Answer:

$$F_c\left[f(x)\right]=\sqrt{\frac{2}{\pi}}\int_0^{\infty} f(x)\cos sx\,dx$$

$$F_c\left[e^{-2x}+2e^{-x}\right]=\sqrt{\frac{2}{\pi}}\int_0^{\infty}\left(e^{-2x}+2e^{-x}\right)\cos sx\,dx$$

$$=\sqrt{\frac{2}{\pi}}\left\{\int_0^{\infty} e^{-2x}\cos sx\,dx+2\int_0^{\infty} e^{-x}\cos sx\,dx\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left\{\frac{2}{s^2+4}+2\frac{1}{s^2+1}\right\}=2\sqrt{\frac{2}{\pi}}\left\{\frac{1}{s^2+4}+\frac{1}{s^2+1}\right\}$$

16. Find the Fourier sine transform of $f(x)=\begin{cases}1, & 0\le x\le 1\\ 0 & x>1\end{cases}$

Answer:

$$F_s\left[f(x)\right]=\sqrt{\frac{2}{\pi}}\int_0^{\infty} f(x)\sin sx\,dx$$

$$F_s\left[f(x)\right]=\sqrt{\frac{2}{\pi}}\left\{\int_0^{1} f(x)\sin sx\,dx+\int_1^{\infty} f(x)\sin sx\,dx\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left\{\int_0^{1} 1\sin sx\,dx+0\right\}=\sqrt{\frac{2}{\pi}}\left[-\frac{\cos sx}{s}\right]_0^1$$

$$=\sqrt{\frac{2}{\pi}}\left[\frac{1-\cos s}{s}\right]$$

17. Obtain the Fourier sine transform of $f(x)=\begin{cases} x, & o<x<1 \\ 2-x, & 1<x<2 \\ 0, & x>2 \end{cases}$.

<u>Answer:</u>

$$F_s\left[f\ x\ \right]=\sqrt{\frac{2}{\pi}}\int_0^{\infty} f\ x\ \sin sxdx$$

$$=\sqrt{\frac{2}{\pi}}\left\{\int_0^1 x\sin sxdx+\int_1^2 2-x\ \sin sxdx\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left\{\left[-x\frac{\cos sx}{s}+\frac{\sin sx}{s^2}\right]_0^1+\left[-\ 2-x\ \frac{\cos sx}{s}-\frac{\sin sx}{s^2}\right]_1^2\right\}$$

$$F_s\left[f\ x\ \right]=\sqrt{\frac{2}{\pi}}\left\{-\frac{\cos s}{s}+\frac{\sin s}{s^2}-\frac{\sin 2s}{s^2}+\frac{\cos s}{s}+\frac{\sin s}{s^2}\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left\{\frac{2\sin s-\sin}{2s\ s^2}\right\}$$

18. Define self reciprocal and give example.

If the transform of f x is equal to f s , then the function f x is called self reciprocal. $e^{-x^2/2}$ *is self reciprocal under Fourier transform.*

19. Find the Fourier cosine Transform of $f(x)=\begin{cases} x & 0<x<\pi \\ 0 & x\geq\pi \end{cases}$

Answer:

$$F_c\left[f\left(x\right)\right]=\sqrt{\frac{2}{\pi}}\int_0^\infty f\left(x\right)\cos sx\,dx \quad =\sqrt{\frac{2}{\pi}}\int_0^\pi x\cos sx\,dx$$

$$=\sqrt{\frac{2}{\pi}}\left[x\left(\frac{\sin sx}{s}\right)+\frac{\cos sx}{s^2}\right]_0^\pi =\sqrt{\frac{2}{\pi}}\left[\frac{\pi}{s}\sin s\pi+\frac{\cos s\pi}{s^2}-\frac{1}{s^2}\right]$$

$$=\sqrt{\frac{2}{\pi}}\left[\frac{\pi s\sin s\pi+\cos s\pi-1}{s^2}\right]$$

20. Find the Fourier sine transform of $\frac{x}{x^2+a^2}$ **.**

Answer:

L et $f\left(x\right)=e^{-ax}$

$$F_s\left[e^{-ax}\right]=\sqrt{\frac{2}{\pi}}\frac{s}{s^2+a^2}$$

Using Inverse formula for Fourier sine transforms

$$e^{-ax}=\sqrt{\frac{2}{\pi}}\int_0^\infty\sqrt{\frac{2}{\pi}}\frac{s}{s^2+a^2}\sin sx\,ds$$

$$(ie)\int_0^\infty\frac{s}{s^2+a^2}\sin sx\,ds \quad =\frac{\pi}{2}e^{-ax},\ a>0$$

Change x and s, we get $\int_0^\infty \frac{x}{x^2+a^2}\sin sx\,dx = \frac{\pi}{2}e^{-as}$

$$F_s\left[\frac{x}{x^2+a^2}\right]=\sqrt{\frac{2}{\pi}}\int_0^\infty \frac{x}{x^2+a^2}\sin sx\,dx$$

$$=\sqrt{\frac{2}{\pi}}\;\frac{\pi}{2}\;e^{-as} \qquad =\sqrt{\frac{\pi}{2}}\;e^{-as}$$

FOURIER TRANSFORM

PART-B

1. (i)Find the Fourier Transform of $f(x)=\begin{cases}1-x^2 & if\ |x|<1\\ 0 & if\ |x|>1\end{cases}$ *and hence*

deduce that (i) $\int_0^\infty \left(\frac{x\cos x-\sin x}{x^3}\right)\cos\left(\frac{x}{2}\right)dx=-\frac{3\pi}{16}$ *(ii)* $\int_0^\infty \left(\frac{\sin x - x\cos x}{x^3}\right)^2 dx=\frac{\pi}{15}$

(ii). Find the Fourier Transform of $f(x)=\begin{cases}a^2-x^2 & |x|<a\\ 0 & |x|>a>0\end{cases}$ *. hence*

deduce that $\int_0^\infty \left(\frac{\sin x - x\cos x}{x^3}\right)dx=\frac{\pi}{4}$

2. Find the Fourier Transform of $f(x)=\begin{cases}1 & if\ |x|<a\\ 0 & if\ |x|>a\end{cases}$ *and hence evaluate*

i) $\int_0^\infty \frac{\sin x}{x}dx$ *ii)* $\int_0^\infty \left(\frac{\sin x}{x}\right)^2 dx$

4. Find Fourier Transform of $f(x)=\begin{cases}1-|x| & if\ |x|<1\\ 0 & if\ |x|>1\end{cases}$ *and hence evaluate*

i) $\int_0^\infty \left(\frac{\sin x}{x}\right)^2 dx$ *ii)* $\int_0^\infty \left(\frac{\sin x}{x}\right)^4 dx$

5. Evaluate *i)* $\int_0^\infty \frac{x^2}{(x^2+a^2)^2}dx$ *ii)* $\int_0^\infty \frac{dx}{(x^2+a^2)^2}$

6 i). Evaluate (a) $\int_0^\infty \frac{dx}{(x^2+a^2)(x^2+b^2)}$ *(b)* $\int_0^\infty \frac{x^2dx}{(x^2+a^2)(x^2+b^2)}$

ii). Evaluate (a) $\int_0^\infty \frac{dx}{(x^2+1)(x^2+4)}$ *(b)* $\int_0^\infty \frac{t^2dt}{(t^2+4)(t^2+9)}$

7. *(i)Find the Fourier sine transform of* $f(x)=\begin{cases}\sin x; when\ o < x < \pi \\ 0 \quad ; when x > \pi\end{cases}$

(ii) *Find the Fourier cosine transform of* $f(x)=\begin{cases}\cos x; when\ o < x < a \\ 0 \quad ; when x > a\end{cases}$

8. (i) Show that Fourier transform $e^{\frac{-x^2}{2}}$ *is* $e^{\frac{-s^2}{2}}$

(ii)Obtain Fourier cosine Transform of $e^{-a^2x^2}$ *and hence find Fourier sine Transform* $xe^{-a^2x^2}$

9. (i) Solve for f(x) from the integral equation $\int_0^\infty f(x)\cos\alpha x dx = e^{-\alpha}$

(ii) Solve for f(x) from the integral equation $\int_0^\infty f(x)\sin tx dx = \begin{cases}1 & ,0 \le t < 1 \\ 2 & ,1 \le t < 2 \\ 0 & ,t \ge 2\end{cases}$

10. (i) Find Fourier sine Transform of e^{-x}*, x>0 and hence deduce that* $\int_0^\infty \frac{x\sin x}{1+x^2}dx$

(ii) Find Fourier cosine and sine Transform of e^{-4x}*, x>0 and hence deduce*

that $(i)\int_0^\infty \frac{\cos 2x}{x^2+16}dx = \frac{\pi}{8}e^{-8}$ $(ii)\int_0^\infty \frac{x\sin 2x}{x^2+16}dx = \frac{\pi}{8}e^{-8}$

11.(i)Find $F_S\left(xe^{-ax}\right) \,\&\, F_c\left(xe^{-ax}\right)$

(ii) Find $F_S\left(\frac{e^{-ax}}{x}\right) \,\&\, F_c\left(\frac{e^{-ax}}{x}\right)$

(iii) *Find the Fourier cosine transform of* $f(x) = e^{-ax}\cos ax$